Sprint Plan

Coinscious: Smart Expense Splitter & Tracker

Document	Sprint Plan
Project	Coinscious: Smart Expense Splitter & Tracker
Course	IT314 Software Engineering
Team	Group 28 (10 members)
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This document sets out how Coinscious is delivered over a one-week setup sprint and four delivery sprints, from 13 October to the final presentation on 30 November 2026. For each sprint it gives the dates, the sprint goal, the user stories, the work of every domain, the testing focus and what is shown at the sprint review. Story IDs (US-xx) come from Final_User Stories.docx (US-36 to US-39 were added by the team on 14 October 2026); task IDs come from docs/backlog.csv.

1. The plan at a glance

Sprint	Theme	Planned dates	Review	Scrum Master	Story points
Sprint 0	Setup and the walking skeleton	13 – 19 Oct	Mon 19 Oct	Ved Bhoraniya	41
Sprint 1	My account and my expenses	20 Oct – 2 Nov	Mon 2 Nov	Harshul Parikh	77 (57 story + 20 setup and testing)
Sprint 2	Groups and money	3 – 16 Nov	Mon 16 Nov	Vishva Vatalia	75 (67 story + 8 setup and testing)
Sprint 3	Smart and automatic	17 – 25 Nov	Wed 25 Nov	Aryan Panara	67 (55 story + 12 setup and testing)
Sprint 4	Ship it	26 – 30 Nov	Mon 30 Nov	Yagnik Arvadiya	32
Total		13 Oct – 30 Nov 2026			**292**

Dates are the planned baseline for 2026. Must work is committed. Stretch items (section 9) are taken into a sprint at sprint planning only when that sprint's Must work is on track, and never after the feature freeze on Wednesday 25 November.

2. Team and roles

Vishwa Patel is the Product Owner. The Scrum Master role rotates each sprint, as shown in the table above. Domain leads: Frontend Riya Soni, Backend Jay Vaghela, Database Vishva Vatalia, Gen-AI Yagnik Arvadiya, Testing Parthiv Panchal.

Member	GitHub	Role	Main responsibilities
Jay Vaghela	@202401237	Backend Engineer · Backend lead	Owns authentication and sessions, the split service and shared-expense API, deployment and security hardening.
Vivek Dodiya	@Dodiya612	Frontend Engineer · Backend · Database · Repo owner	Owns the register and login flow, the group, invite, join and leave screens and API, the recurring, notification and group-settings screens, and repo administration.

Member	GitHub	Role	Main responsibilities
Ved Bhoraniya	@VEd-b4	Frontend Engineer · Database · Gen-AI	Owns the expense screens, dashboard and analytics, the bill-scan and AI screens, web compatibility and the money utilities.
Riya Soni	@RiyaSoni-20	Frontend Engineer (UI/UX lead) · Backend	Owns the design system and screen designs, the split editor, settle-up and budgets screens, the categories API, and the APK and website releases.
Yagnik Arvadiya	@202401252yagnik	Backend · Gen-AI Engineer · Gen-AI lead	Owns the expenses API, balances and settlements, AI categorization, the AI chat and the AI quotas.
Vishwa Patel	@VishwaPatel07	Database Engineer · Product Owner	Owns and orders the backlog and accepts stories at review; owns the expense search queries, the group activity log, analytics, notification triggers and demo data.
Vishva Vatalia	@Vishva1240	Database Engineer · Database lead	Owns migrations, the my-spending query, budgets, the group roles API and backups.
Harshul Parikh	@Parikh-Harshul	Database Engineer	Owns the group schema and seed data, the cloud database, recurring expenses and the performance review.
Aryan Panara	@Aryan99099	Gen-AI · Testing Engineer	Owns AI budget goal suggestions, the AI evaluation, mobile CI and the property-based tests.
Parthiv Panchal	@202401259	Testing · Gen-AI Engineer · Testing lead	Owns the backend test harness and API test suites, bill-scan AI and bug triage.

Vishwa Patel and Vishva Vatalia are two different members.

3. How the team works

Scrum events

- **Sprint planning.** On the first day of each sprint, the team confirms the goal, the stories, the tasks, one owner per task, the points and each member’s availability.
- **Standup.** A short written update in the team channel on Monday, Wednesday and Friday by 22:00: what was done, what is next, what is blocked, and links to pull requests.
- **Sync call.** 20 minutes every Thursday for blockers, integration and API contract changes.
- **Backlog refinement.** In the middle of each sprint, the Product Owner and the domain leads prepare the stories for the next sprint.
- **Sprint review.** On the last day of the sprint: a live demonstration on a phone and in a browser. The Product Owner accepts each story against its acceptance criteria, and each member explains one of their pull requests in 2 minutes.
- **Retrospective.** After each review, the team records what to start, stop and continue, with two action items and owners.

Types of work

Type	Meaning
Build	Code, tests and documentation, merged into the repository through a reviewed pull request.
Screen design	Screens are designed first (wireframes, then high-fidelity Figma designs) and reviewed by the team; Build tasks then build them and connect them to the API.
Research and process	Work done by hand: spikes, labelling, device and browser testing sessions, and the presentation and viva preparation.

Definition of done

- The acceptance criteria of the story pass, each with an automated or manual test case.
- The work is merged into main through a pull request that another member has approved.
- Continuous integration (lint, type checks and tests) passes on the pull request.
- New backend logic has unit tests; new endpoints have API tests, including an authorization test.
- Endpoints appear in Swagger with examples, and docs/openapi.json is regenerated.
- Screens handle loading, empty and error states and are tested on a real Android phone and in a browser.
- Database changes come with an Alembic migration that upgrades and downgrades cleanly.
- No API key, password or other secret is committed.
- The pull request explains what was done and why, and its author can explain every line.
- Work is committed and merged under the name of the member who did it.

4. Sprint 0: Setup and the walking skeleton

Planned dates	13 – 19 October 2026
Sprint review	Monday 19 October 2026
Scrum Master	Ved Bhoraniya
Story points	41
Sprint goal	Every member can run the full stack locally, and the app shows Backend OK · Database OK on a real phone and in a browser, from a FastAPI endpoint that queries PostgreSQL. CI is green and the backlog is on the GitHub board.

User stories in this sprint

Setup sprint: no user stories are delivered. Its tasks set up the repository, CI, the walking skeleton and the first AI spikes.

Work by domain

Frontend

Task	Story	Type	Owner	Done when
S0-FE-01 Expo app scaffold (Android + web) with Expo Router and API client npx create-expo-app@latest mobile (TypeScript + Expo Router template), pin the SDK, reset example screens, add ESLint + Prettier, src/ layout, EXPO_PUBLIC_API_URL config, src/api/client.ts, and a Home screen that calls /api/v1/health and shows Backend and DB status. Ensure web support is installed (react-dom, react-native-web, @expo/metro-runtime via npx expo install). *Starts after: S0-OPS-01*	—	Build	Vivek Dodiya	App runs in Expo Go on an Android phone AND in Chrome via npx expo start --web; status shows OK/ERROR on both; npm run lint and npx tsc --noEmit pass.
S0-FE-02 Design tokens, theme and low-fidelity wireframes Create src/theme/tokens.ts (colors incl. success/warning/danger for budgets, spacing, typography, radius) and a ThemeProvider. Produce low-fi wireframes for the screens of US-01, US-02, US-07, US-13, US-17, US-20 and US-25; save links/images in docs/design/. *Starts after: S0-FE-01*	—	Screen design	Riya Soni	Tokens used by the health screen; wireframes reviewed by Vivek and Ved; files in docs/design/.
S0-FE-03 Money and date utilities (TypeScript) with tests Create src/lib/money.ts: parseAmountToPaise(string) without floating point, formatINR(paise) with Indian digit grouping, allocate(total, weights) matching the backend rule; src/lib/dates.ts helpers. Jest tests. *Starts after: S0-FE-01*	—	Build	Ved Bhoraniya	Tests pass in CI; inputs like 0.1 and 0.2 produce exact paise; allocate matches the Python version on the same test cases.

Task	Story	Type	Owner	Done when
S0-INT-01 Hello-world vertical slice on a real phone and in a browser Run the backend on 0.0.0.0:8000 with Postgres, set EXPO_PUBLIC_API_URL to the laptop LAN IP, open the app in Expo Go on a phone on the same Wi-Fi and in Chrome, verify the health screen shows DB OK on both. Document steps and the Windows firewall rule in docs/setup/phone-testing.md. *Starts after: S0-FE-01; S0-BE-01; S0-DB-02*	—	Build	Vivek Dodiya	Screen recording of phone and browser showing Backend OK / DB OK attached to the issue; guide reproduced by one other member.

Backend

Task	Story	Type	Owner	Done when
S0-BE-01 FastAPI skeleton with config and /health Create backend/app with main.py (create_app, CORS from env), core/config.py using pydantic-settings, api/v1 router, GET /api/v1/health returning status and db check (SELECT 1). requirements.txt, requirements-dev.txt, pyproject.toml with ruff and pytest config. *Starts after: S0-OPS-01*	—	Build	Jay Vaghela	uvicorn app.main:app --reload starts; /docs shows the endpoint; /api/v1/health returns {status: ok, db: ok} when Postgres runs and HTTP 503 with db: error when it does not; ruff check passes.

Database

Task	Story	Type	Owner	Done when
S0-DB-01 Docker Compose PostgreSQL and local DB guide Add docker-compose.yml with a postgres:16 service (named volume, healthcheck, port 5432, database expense_dev) and an optional pgAdmin profile. Write docs/setup/database.md covering Docker Desktop on Windows and the native PostgreSQL installer alternative, plus connecting with pgAdmin/DBeaver. *Starts after: S0-OPS-01*	—	Build	Vishwa Patel	docker compose up -d db starts a healthy Postgres; two teammates on Windows follow the guide successfully; connection string documented in backend/.env.example.
S0-DB-02 SQLAlchemy session and Alembic setup Add app/db/base.py (DeclarativeBase), app/db/session.py (engine, SessionLocal, get_db dependency), alembic init with env.py reading DATABASE_URL from settings and importing all models, first empty revision, commands documented. *Starts after: S0-BE-01; S0-DB-01*	—	Build	Vishva Vatalia	alembic upgrade head runs on a fresh DB; the health endpoint uses get_db; autogenerate detects a test model; instructions in docs/setup/database.md.
S0-DB-03 Finalize ER diagram and seed script skeleton Review the schema in docs/ARCHITECTURE.md against the acceptance criteria in Final_User Stories.docx with the DB team; record decisions. Create backend/scripts/seed.py skeleton that inserts the default system categories (Food, Groceries, Transport, Shopping, Bills and Utilities, Rent, Entertainment, Health, Education, Travel, Other) idempotently. *Starts after: S0-DB-02*	—	Build	Harshul Parikh	ER diagram updated via a PR approved by Vishwa and Vishva; seed.py can run twice without duplicates (once the categories table exists in Sprint 1).

Task	Story	Type	Owner	Done when
S0-DB-04 Money utilities (Python) with tests Create app/core/money.py: parse_amount(str) -> paise int without floats (accepts 123, 123.4, 123.45; rejects negatives, more than 2 decimals, letters), format_inr(paise), allocate(total, weights) that distributes leftover paise deterministically (used by equal, percentage and parts splits). Unit tests for edge cases. *Starts after: S0-BE-01*	—	Build	Ved Bhoraniya	pytest passes; allocate(10000, [1,1,1]) == [3334, 3333, 3333] and always sums to total; no float used anywhere in the module.

Gen-AI

Task	Story	Type	Owner	Done when
S0-AI-01 LLM provider abstraction and spike Create backend/app/ai/provider.py with an LLMProvider protocol (generate_json(prompt, schema, image=None, timeout)), GeminiProvider (google-genai SDK, key from env) and FakeProvider (canned responses for tests). Spike script categorizes 5 sample descriptions and prints validated JSON. Record the current model name, free-tier limits and data-use terms. *Starts after: S0-BE-01*	—	Build	Yagnik Arvadiya	AI_PROVIDER=fake works with no key; AI_PROVIDER=gemini returns schema-valid JSON for 5 samples; no key committed; findings in docs/ai/provider-notes.md.
S0-AI-02 Bill extraction spike Using the provider from S0-AI-01, test extraction of merchant, total, date and items from 5 real bill photos (shop bills, restaurant bills, UPI screenshots). Record accuracy and failure modes. *Starts after: S0-AI-01*	US-16	Research and process	Parthiv Panchal	docs/ai/receipt-spike.md lists per-bill results and a recommendation.
S0-AI-03 Golden datasets for AI evaluation Create docs/ai/datasets/categorize_golden.json with at least 40 labelled expense descriptions (English and Hinglish, merchants like Swiggy, Zomato, Uber, DMart) and collect 10 bill photos without sensitive data in a private shared folder. *Starts at the start of the sprint*	US-16, US-26	Research and process	Aryan Panara	40+ labelled rows covering every default category; reviewed by Yagnik.

Testing

Task	Story	Type	Owner	Done when
S0-QA-01 Backend test harness and backend CI tests/conftest.py with a separate test database (schema created via Alembic), TestClient fixture and a first test for /health. .github/workflows/backend.yml: Python 3.12, pip install, ruff check, ruff format --check, Postgres service container, alembic upgrade head, pytest. *Starts after: S0-BE-01; S0-DB-02*	—	Build	Parthiv Panchal	CI runs on PRs touching backend/** and is green; a failing test turns the PR check red; documented in docs/testing/test-plan.md.

Task	Story	Type	Owner	Done when
S0-QA-02 Mobile CI and first component test Configure jest-expo and @testing-library/react-native, write a test for the health status component (OK and ERROR with mocked API). .github/workflows/mobile.yml: Node 22 LTS, npm ci, lint, tsc --noEmit, jest. *Starts after: S0-FE-01*	—	Build	Aryan Panara	CI runs on PRs touching mobile/** and is green; the test fails when the status text is broken.
S0-QA-03 Test plan and test case / bug templates Write docs/testing/test-plan.md (scope, levels, tools, device and browser matrix, entry/exit criteria) and the docs/testing/test-cases.md template where every case references a story ID from Final_User Stories.docx (US-01..US-35). Bug template fields: steps, expected, actual, severity S1-S4, device/browser, build. *Starts after: S0-OPS-03*	—	Build	Aryan Panara	Reviewed by Parthiv and Jay; bug template live in .github/ISSUE_TEMPLATE/bug.yml.

DevOps

Task	Story	Type	Owner	Done when
S0-OPS-01 Monorepo scaffold and repo hygiene Create /mobile, /backend, /docs, /.github folders in the existing repo. Add root .gitignore (Python, Node, Expo, .venv, node_modules, .expo, dist, .env, *.apk), .gitattributes (eol=lf), .editorconfig, backend/.env.example and mobile/.env.example. Update README: tech stack, development status, product name Coinscious, link to CONTRIBUTING.md. *Starts at the start of the sprint*	—	Build	Jay Vaghela	Folders exist on main; .env files are ignored (git check-ignore passes); README lists the fixed tech stack; no secrets in repo.
S0-OPS-02 Branch protection and merge settings Protect main: require PR, 1 approving review, passing status checks (added once CI exists), no force-push, no deletion. Enable Create-a-merge-commit (keeps every individual commit and its author in history). Vivek does this as the repo owner. *Starts after: S0-OPS-01*	—	Build	Vivek Dodiya	Direct push to main is rejected; a PR without approval cannot be merged; merge-commit option enabled.
S0-OPS-03 Issue/PR templates, labels, Projects board, CONTRIBUTING.md Add .github/ISSUE_TEMPLATE (feature.yml with user-story and acceptance-criteria fields, bug.yml, task.yml) and pull_request_template.md. Create labels (domain:*, type:*, priority:*, sprint:*, severity:*, status:blocked) and one label per epic (epic:EP-01 to epic:EP-09, ROADMAP.md §2.1). Create the GitHub Project board (Backlog, Sprint Ready, In Progress, In Review, Done) and milestones Sprint 0-4 with due dates. Write CONTRIBUTING.md. *Starts after: S0-OPS-01*	—	Build	Riya Soni	Templates appear when opening an issue/PR; all labels and milestones exist; board visible to all 10 members; CONTRIBUTING.md linked from README.

Task	Story	Type	Owner	Done when
S0-OPS-04 Import backlog into GitHub Issues Write backend/scripts/import_backlog.py that reads docs/backlog.csv and calls the gh CLI to create one issue per row: title [ID] Title, body = description + acceptance criteria + dependencies, labels for domain/sprint/priority, milestone = sprint, assignees mapped from names to GitHub handles (ROADMAP.md team directory). Support --dry-run. *Starts after: S0-OPS-03*	—	Build	Vishwa Patel	Dry run prints all issues; real run creates Sprint 0 and 1 issues without duplicates; issues added to the Project board.

Other planned work

Personal setup by every member: accept the repo invite, set git user.email to the GitHub email, install the tools and Expo Go. Kickoff call on Tuesday 13 or Wednesday 14 October, first written standup on Wednesday 14 October, and a contribution log for each member in docs/contributions/. Ved Bhoraniya commits Final_User Stories.docx, the SRS and the activity diagram into docs/requirements/. Integration check on Sunday 18 October: Harshul Parikh runs the walking skeleton using only the written guides.

Testing focus

CI for both apps, the backend test harness, the test plan and the golden dataset.

Sprint review

- **Demonstration.** The phone and the browser show Backend OK / DB OK; both CI checks are green on a pull request; the board is populated; Yagnik Arvadiya shows AI categorizing 5 sample expenses.

Exit criteria

Sprint 0 is complete only when all of the following are true:

1. The walking skeleton is reproduced from the guides by a member who did not build it
2. main is protected
3. Every one of the 10 members has at least one merged pull request

5. Sprint 1: My account and my expenses

Planned dates	20 October – 2 November 2026
Sprint review	Monday 2 November 2026
Scrum Master	Harshul Parikh
Story points	77 (57 story + 20 setup and testing)
Sprint goal	A user can register, log in (with Remember me) and add, edit, delete (with undo) and browse personal expenses, with AI category suggestions, on a phone and in a browser. The group and activity foundations are ready for Sprint 2.

User stories in this sprint

ID	Title	Priority	Points	Owner
US-01	Registration	Must	5	Jay Vaghela, with Vivek Dodiya, Aryan Panara and Vishva Vatalia
US-02	Login	Must	5	Jay Vaghela, with Vivek Dodiya, Aryan Panara and Vishva Vatalia
US-03	Password Visibility Toggle	Must	1	Vivek Dodiya
US-05	Remember Me and Multi-Device Sync	Must	4	Jay Vaghela, with Vivek Dodiya, Aryan Panara and Vishva Vatalia
US-07	Add Expense	Must	10	Vishwa Patel, with Ved Bhoraniya, Riya Soni, Yagnik Arvadiya and Parthiv Panchal
US-08	Delete Expense	Must	3	Yagnik Arvadiya, with Ved Bhoraniya and Parthiv Panchal
US-09	Edit Expense	Must	3	Yagnik Arvadiya, with Ved Bhoraniya and Parthiv Panchal
US-12	Total Spending Over Time (groundwork)	Must	1	Vishva Vatalia
US-13	Recent Expenses (groundwork)	Must	3	Ved Bhoraniya, with Vishwa Patel
US-15	Search and Filter Transactions (groundwork)	Must	2	Vishwa Patel, with Parthiv Panchal
US-16	Scan Bill to Create Expense (groundwork)	Must	3	Parthiv Panchal
US-17	Create Group and Invite Members	Must	4	Vivek Dodiya, with Harshul Parikh
US-21	Manage Group Details and Roles (groundwork)	Must	1	Harshul Parikh
US-22	Leave Group (groundwork)	Must	1	Harshul Parikh
US-23	Edit or Delete Group Expense	Must	3	Vishwa Patel, with Harshul Parikh
US-25	AI Goal Setting (groundwork)	Must	1	Vishva Vatalia
US-26	AI Category Suggestion	Must	7	Yagnik Arvadiya, with Aryan Panara and Ved Bhoraniya
	Total		**57**	

Points are the backlog estimates of this sprint's tasks for each story; a task that serves several stories is split between them. "Groundwork" stories are delivered in a later sprint. The other 20 points are setup and testing tasks with no story.

Work by domain

Frontend

Task	Story	Type	Owner	Done when
S1-FE-01 Shared component library src/components: Button, TextField, PasswordField with eye toggle (masked by default, US-03), AmountInput, Select, Card, ListItem (supports swipe actions on Android), Screen wrapper (safe area, max width on web), Loading/Empty/Error states, Badge, ConfirmDialog, Snackbar with action (used for Undo, US-08). A /dev/components screen shows all components. *Starts after: S0-FE-02*	—	Build	Riya Soni	Components use tokens only; accessible labels set; PasswordField toggles on Android and web; reviewed by Vivek and Ved.
S1-FE-02 Hi-fi designs for Sprint 2 screens High-fidelity Figma designs for groups list/detail, invites and join, add shared expense with split editor, balances/settle-up, home dashboard, analytics and search/filter. *Starts after: S0-FE-02*	—	Screen design	Riya Soni	Designs linked in docs/design/README.md; reviewed at the Sprint 1 review.
S1-FE-03 Auth flow: register, login, remember me, token storage, sync Register screen (email, password, confirm password; Account already exists. Please log in with a link to login; Already have an account? Login link; redirect to dashboard on success). Login screen (email, password, Remember me checkbox, errors). src/lib/tokenStorage.ts: SecureStore on Android, localStorage on web, memory only when Remember me is unchecked. API client attaches the token and refreshes once on 401. Route guards for (auth) and (app) groups. TanStack Query set to refetch on focus and poll every 30 s on open lists (multi-device sync). *Starts after: S1-FE-01; S1-BE-01*	US-01, US-02, US-03, US-05	Build	Vivek Dodiya	Every US-01 and US-02 acceptance criterion passes on Android and web; with Remember me the user stays logged in after restart, without it they do not; a change made on device A appears on device B within about 30 s.
S1-FE-04 Expense history screen QueryProvider setup, Expenses tab: list grouped by date with amount, category, payment mode and notes preview, tap for details, pull-to-refresh, infinite scroll, loading/empty/error states. Mock data first, then the real API. *Starts after: S1-FE-01; S1-BE-03*	US-07, US-13	Build	Ved Bhoraniya	New expense appears instantly after saving; empty state for a new user; pagination loads the next page; works on phone and web.
S1-FE-05 Add / edit / delete expense Form: amount, description, category picker, date, payment mode, optional notes, merchant; zod validation mirroring the backend; edit mode pre-filled with a Cancel button that discards changes; delete asks for confirmation, then hides the item and shows a 5-second Undo snackbar before sending DELETE. *Starts after: S1-FE-01; S1-BE-03; S1-BE-04*	US-07, US-08, US-09	Build	Ved Bhoraniya	Cannot save an invalid amount; server validation errors shown on fields; Undo restores the expense and no DELETE is sent; edits appear immediately in history and charts.

Task	Story	Type	Owner	Done when
S1-FE-06 AI category suggestion in expense form When the user types a description (debounced 600 ms), call /ai/categorize and show the suggested category as a chip with confidence; the user taps to accept or picks another category; it never overrides a category the user already chose. *Starts after: S1-FE-05; S1-AI-01*	US-26	Build	Ved Bhoraniya	Suggestion appears while typing; manual choice always wins; the form works normally if the AI call fails or is slow.
S1-FE-07 Web compatibility baseline Run every Sprint 1 screen in the browser, fix platform issues (Platform.select or .web.tsx files), check web-safe date picker and dialogs, provide visible Edit/Delete buttons on web where Android uses swipe. Write docs/setup/web.md with known differences. *Starts after: S1-FE-03; S1-FE-05*	—	Build	Ved Bhoraniya	Register, login, expense list and expense form work in Chrome at desktop and phone widths; no native-only module crashes on web.

Backend

Task	Story	Type	Owner	Done when
S1-BE-01 Authentication: register, login, remember me, refresh, logout core/security.py with pwdlib (argon2) and PyJWT access tokens (30 min). /auth/register (duplicate email returns 409 code ACCOUNT_EXISTS so the app can link to login), /auth/login with remember_me and device_name (refresh token 30 days if remember_me, else 24 h; one refresh token per device so concurrent logins work), /auth/refresh with rotation, /auth/logout, dependency get_current_user. Publish a stub get_current_user on day 1. *Starts after: S1-DB-01*	US-01, US-02, US-05	Build	Jay Vaghela	Passwords never stored or logged in plain text; wrong credentials return 401 with a generic message; two devices can be logged in at once; a rotated refresh token cannot be reused; documented in /docs.
S1-BE-02 Error format, /users/me, OpenAPI export Global exception handlers producing {error: {code, message, details}} for validation, 401, 403, 404, 409, 429, 500. GET/PATCH /users/me. scripts/export_openapi.py writes docs/openapi.json; add it to the PR checklist. *Starts after: S1-BE-01*	—	Build	Jay Vaghela	All errors follow the shape (tested); openapi.json regenerated and committed.
S1-BE-03 Personal expenses CRUD API Pydantic schemas (amount_minor > 0, description 1-200 chars, notes up to 500, payment_mode from the allowed list, expense_date not more than 1 day in the future, category visible to the user, optional receipt_id). GET/POST /expenses, GET/PATCH/DELETE /expenses/{id} using repositories/expenses.py; owner-only (404 otherwise); edits are re-validated like creates. Contract stub PR on day 2. *Starts after: S1-DB-02; S1-DB-05; S1-BE-01*	US-07, US-08, US-09	Build	Yagnik Arvadiya	All endpoints in /docs with examples; invalid amount returns 422; another user's expense returns 404; list supports all US-15 filters and pagination.

Task	Story	Type	Owner	Done when
S1-BE-04 Categories API GET /categories (system + own), POST/PATCH/DELETE own custom categories; duplicate name returns 409; deleting a category in use returns 409 with a message. *Starts after: S1-DB-02; S1-BE-01*	US-07	Build	Riya Soni	System categories cannot be edited or deleted; custom ones visible only to their owner; tests pass.
S1-BE-05 Groups API and role dependencies GET/POST /groups (creator becomes admin), GET /groups/{id} with members and roles. Dependencies require_group_member and require_group_admin reused by all later group endpoints. *Starts after: S1-DB-03; S1-BE-01*	US-17	Build	Vivek Dodiya	Non-members get 404 on group endpoints; creator is admin; tests pass.

Database

Task	Story	Type	Owner	Done when
S1-DB-01 Models and migration: users, refresh_tokens users (unique lower(email), role check, is_active) and refresh_tokens (token_hash unique, device_name, remember_me, created_at, last_used_at, expires_at, revoked_at). Must be merged by Thu 22 Oct (blocks auth). *Starts after: S0-DB-02*	US-01, US-02, US-05	Build	Vishva Vatalia	alembic upgrade and downgrade both work; unique email enforced case-insensitively; reviewed by Jay.
S1-DB-02 Models and migration: categories, expenses, expense_payers, expense_splits Tables per ARCHITECTURE.md: expenses with payment_mode and notes, CHECK amount_minor > 0, split_type check (none, equal, exact, percentage, parts), personal-vs-group check, indexes (created_by, expense_date) and (group_id, expense_date), unique (recurring_rule_id, expense_date); expense_payers; expense_splits with percent_bp and parts. Merge by Thu 22 Oct. *Starts after: S0-DB-02*	US-07	Build	Vishwa Patel	Migration applies on an empty DB; amount 0 or negative is rejected; invalid payment_mode rejected; indexes visible in psql.
S1-DB-03 Models and migration: groups, group_members, group_invites, group_activity groups (name, description), group_members (composite PK, role admin/member, joined_at, left_at), group_invites (unique code, invited_email, status, expires_at, partial unique pending invite per email), group_activity (action, details JSONB). Merge by Fri 23 Oct. *Starts after: S0-DB-02*	US-17, US-21, US-22, US-23	Build	Harshul Parikh	Duplicate membership rejected; duplicate pending invite for the same email rejected; deleting a group cascades to members, invites and activity.
S1-DB-04 Seed script: categories, demo users, sample expenses Extend scripts/seed.py: default categories, 5 demo users (password from env, documented), 30 personal expenses over the last 2 months across categories and payment modes. Idempotent; --reset clears demo data. *Starts after: S1-DB-01; S1-DB-02; S1-DB-03*	—	Build	Harshul Parikh	python -m scripts.seed runs twice without duplicates; the frontend team can log in as demo users; documented in docs/setup/database.md.

Task	Story	Type	Owner	Done when
S1-DB-05 Expense query module: search, filters, pagination repositories/expenses.py: list_expenses(user_id, scope personal|all, q, date_from, date_to, category_id, min_amount, max_amount, payment_mode, limit, offset) returning items + total, newest first. q matches description, notes, merchant and category name case-insensitively. scope=all also returns group expenses where the user is a payer or participant. get_by_id scoped to the owner. *Starts after: S1-DB-02*	US-13, US-15	Build	Vishwa Patel	All filters combine correctly; total count correct; tests cover each filter and the empty result.
S1-DB-06 DB constraint tests and migration check in CI Tests that the DB rejects invalid data (amount <= 0, duplicate email, invalid split_type, invalid payment_mode, invalid group role). CI step: upgrade head, downgrade base, upgrade again; fail if alembic heads shows more than one head. *Starts after: S1-DB-01; S1-DB-02; S0-QA-01*	—	Build	Vishva Vatalia	CI fails on a branch with two migration heads; constraint tests pass.
S1-DB-07 My-spending query helper repositories/spending.py: user_spending(user_id, date_from, date_to, category_id=None) = personal expenses created by the user + the user's share_minor in group expenses. Used by analytics, budgets and AI. Tests with personal and group fixtures. *Starts after: S1-DB-02*	US-12, US-25	Build	Vishva Vatalia	Returns integer paise; tests cover personal only, group only, mixed and empty ranges.
S1-DB-08 DB reset helper scripts/reset_db.py drops and recreates the dev schema via Alembic and re-seeds; refuses to run if ENV=production. *Starts after: S1-DB-04*	—	Build	Harshul Parikh	One command restores a clean seeded dev DB; refuses on production.
S1-DB-09 Group activity log helper and endpoint services/activity.py record(group_id, actor_id, action, details) used by every group-changing service, and GET /groups/{id}/activity (members only, paginated, newest first). *Starts after: S1-DB-03; S1-BE-05*	US-23	Build	Vishwa Patel	Activity rows written in the same transaction as the change; non-members get 404; tests pass.

Gen-AI

Task	Story	Type	Owner	Done when
S1-AI-01 Expense auto-categorization endpoint POST /ai/categorize: prompt with the user's allowed category names, structured JSON output {category, confidence}, Pydantic validation, mapping to category_id, cache by normalized description per user, fallback to rule-based categorizer on timeout (8 s) or invalid output. Response includes source ai|rules|cache. *Starts after: S0-AI-01; S1-BE-01; S1-DB-02*	US-26	Build	Yagnik Arvadiya	Never returns a category outside the user's list; works with AI_PROVIDER=fake; returns a rules result when the provider fails; unit tests with FakeProvider.

Task	Story	Type	Owner	Done when
S1-AI-02 Rule-based fallback categorizer and eval script ai/fallback_rules.py keyword map (Swiggy/Zomato to Food, Uber/Ola/metro to Transport, etc.). scripts/eval_categorize.py runs the golden dataset through rules and through the AI provider and prints accuracy and confusion pairs (manual run, not CI). *Starts after: S0-AI-03; S0-AI-01*	US-26	Build	Aryan Panara	Rules and AI accuracy reported in docs/ai/eval-sprint1.md.
S1-AI-03 Bill extraction service (no endpoint yet) ai/receipt.py: ReceiptExtraction schema (merchant, total_minor, currency, date, items[name, quantity, amount_minor], suggested_category, warnings[]), prompt template in ai/prompts/receipt.md, image downscale with Pillow (max 1600 px), sanity checks (total > 0, date not in the future, items sum vs total warning). Unit tests with FakeProvider fixtures. *Starts after: S0-AI-01; S0-AI-02*	US-16	Build	Parthiv Panchal	Invalid or partial AI output never raises to the caller; returns status failed with a reason; tests cover good, partial, garbage and prompt-injection text.

Testing

Task	Story	Type	Owner	Done when
S1-QA-01 API tests: authentication tests/api/test_auth.py: register success, duplicate email (case-insensitive, ACCOUNT_EXISTS), weak password, login success/failure, remember_me token lifetimes, two concurrent sessions, protected endpoint with missing/expired token, refresh rotation and reuse rejection, logout. *Starts after: S1-BE-01*	US-01, US-02, US-05	Build	Aryan Panara	At least 14 test cases; all pass in CI; bugs filed as issues.
S1-QA-02 API tests: expenses and categories CRUD happy paths, validation errors (amount, payment mode, future date), every US-15 filter and pagination, another user's data returns 404, category in use cannot be deleted. *Starts after: S1-BE-03; S1-BE-04*	US-07, US-08, US-09, US-15	Build	Parthiv Panchal	At least 18 test cases; all pass in CI; authorization tests included.
S1-QA-03 Sprint 1 manual test cases and device session Write manual test cases for US-01, US-02, US-03, US-05, US-07, US-08, US-09 and US-26 in docs/testing/test-cases.md (one per acceptance criterion), run them on 2 Android phones and Chrome before the sprint review, file bugs. *Starts after: S1-FE-03; S1-FE-05*	—	Research and process	Aryan Panara	Results table committed (pass/fail per case per device/browser).
S1-QA-04 Frontend component tests: auth and expense form Jest + Testing Library tests: register form (confirm password mismatch, account-exists link), PasswordField toggle, add-expense validation, Undo snackbar cancels the delete call. *Starts after: S1-FE-03; S1-FE-05*	—	Build	Parthiv Panchal	Tests pass in mobile CI; at least 10 assertions.

Other planned work

Key dates: Wednesday 21 October, the stub get_current_user and the contract pull requests for expenses and groups; Thursday 22 October, the users and expenses migrations and the core components; Friday 23 October, the groups migration. Screens are built against mock data until the real endpoints are merged.

Testing focus

Auth and expense API tests, form component tests, and the first device and browser session.

Sprint review

- **Demonstration.** On a phone: register (an existing email shows “Account already exists. Please log in”), log in with Remember me and toggle the password eye; add 3 expenses with payment modes and accept one AI category suggestion; edit one and cancel another edit; delete one and undo; reopen the app and stay logged in; open the website as the same user and see an expense added there appear on the phone within about 30 s.

Exit criteria

Sprint 1 is complete only when all of the following are true:

1. Every task meets the definition of done
2. The test cases for US-01, US-02, US-03, US-05 (AC1–2), US-07 (AC1–3), US-08, US-09 and US-26 pass on 2 phones and in Chrome

6. Sprint 2: Groups and money

Planned dates	3 – 16 November 2026
Sprint review	Monday 16 November 2026
Scrum Master	Vishva Vatalia
Story points	75 (67 story + 8 setup and testing)
Sprint goal	Users can create groups, invite and join by link, code or email, add shared expenses split equally, by amounts, percentages or parts, see simple balances and settle up (fully or partially), leave a group, see the group history, search their own expenses and see spending over time. The backend runs on HTTPS staging.

User stories in this sprint

ID	Title	Priority	Points	Owner
US-10	Track Money Owed	Must	5	Riya Soni, with Yagnik Arvadiya
US-12	Total Spending Over Time	Must	10	Vishwa Patel, with Ved Bhoraniya
US-13	Recent Expenses	Must	3	Ved Bhoraniya
US-14	Recurring Expenses (groundwork)	Must	2	Harshul Parikh
US-15	Search and Filter Transactions	Must	2	Ved Bhoraniya
US-16	Scan Bill to Create Expense	Must	5	Parthiv Panchal, with Harshul Parikh
US-17	Create Group and Invite Members	Must	4	Vivek Dodiya, with Parthiv Panchal
US-18	Add Shared Expense	Must	4	Jay Vaghela, with Riya Soni and Parthiv Panchal
US-19	Unequal Split	Must	5	Jay Vaghela, with Aryan Panara and Riya Soni
US-20	View Balances	Must	7	Yagnik Arvadiya, with Riya Soni, Aryan Panara, Vishva Vatalia and Parthiv Panchal
US-22	Leave Group	Must	4	Vivek Dodiya, with Parthiv Panchal
US-23	Edit or Delete Group Expense	Must	3	Vivek Dodiya, with Jay Vaghela and Parthiv Panchal
US-25	AI Goal Setting	Must	11	Vishva Vatalia, with Aryan Panara and Parthiv Panchal
US-34	Data Integrity	Must	2	Harshul Parikh
	Total		**67**	

Points are the backlog estimates of this sprint's tasks for each story; a task that serves several stories is split between them. "Groundwork" stories are delivered in a later sprint. The other 8 points are setup and testing tasks with no story.

Work by domain

Frontend

Task	Story	Type	Owner	Done when
S2-FE-01 Groups screens: list, create, invite, join, leave, activity Groups tab with my net balance per group; create group (name, description); invite screen that shows the link/code with a Share button (phone share sheet: WhatsApp/SMS) and an invite-by-email field; join by link (deep link /join/[code]) or by typing a code; pending invites with Accept/Decline; group detail with members, roles, activity history; Leave group with confirmation and an unsettled-balance warning; admin Remove member with the same balance warning. *Starts after: S2-BE-06; S1-FE-01*	US-17, US-22, US-23	Build	Vivek Dodiya	Every US-17 and US-22 acceptance criterion passes on Android and web except email delivery (in-app only, see ROADMAP.md §3.5 D6); admin-only actions hidden for members.
S2-FE-02 Add shared expense with split editor Group expense form: amount, description, category, payer (member picker), participants (default all members, equal split), split tabs Equal/Exact/Percent/Parts with live remaining-amount indicator, optional notes; Save disabled until valid. Add a Split with a group switch on the personal Add Expense form that switches to this form. Preview uses the same allocate rule as the backend. *Starts after: S2-BE-02; S1-FE-01*	US-10, US-18, US-19	Build	Riya Soni	Cannot save when exact amounts, percentages or parts are invalid; preview equals the server result; the new expense appears instantly in the group list.
S2-FE-03 Balances and settle-up screen Each member's net balance in simple words (You owe Jay ₹300 / Riya owes you ₹100), simplified transfer list, Settle up button prefilled from a transfer with an editable amount (partial allowed), Remind button next to members who owe you, settlement history. *Starts after: S2-BE-04; S1-FE-01*	US-10, US-20	Build	Riya Soni	Balances refresh right after a settlement; amounts match the API; empty state when everything is settled.
S2-FE-07 Hi-fi designs for Sprint 3 screens Figma designs for bill scan and review, budgets with AI goal suggestions, recurring expenses, notifications, Ask AI chat, group settings and roles, change password. *Starts after: S1-FE-02*	—	Screen design	Riya Soni	Designs linked in docs/design/README.md before Sprint 3 planning.
S2-FE-05 Home dashboard, recent expenses and analytics screens Home: this month's total vs last month, You owe / You are owed, and the 10 most recent expenses, personal and group (date, amount, category, notes; tap for details; swipe to edit/delete on Android, buttons on web; Edit/Delete only offered where the user is allowed). Analytics screen: cumulative total, time-range filter, spend over time chart and category breakdown using react-native-gifted-charts. *Starts after: S2-DB-03; S2-BE-04; S1-FE-04*	US-12, US-13	Build	Ved Bhoraniya	Charts render on Android and web; values match the analytics API; editing or deleting an expense updates the dashboard and charts immediately.

Task	Story	Type	Owner	Done when
S2-FE-06 Search and filter transactions Search bar (description, notes, merchant, category), filters for date range, category, amount range and payment mode, combinable, with one-tap Clear all, and a No results found empty state. *Starts after: S1-FE-04; S1-DB-05*	US-15	Build	Ved Bhoraniya	Each filter and combinations work against the API; Clear all resets everything; empty state text shown when nothing matches.

Backend

Task	Story	Type	Owner	Done when
S2-BE-05 Budgets API GET/POST /budgets (manual or accepting an AI suggestion), PATCH/DELETE /budgets/{id}, GET /budgets/status wired to S2-DB-02. Validation: limit > 0, threshold 1-100, category visible to the user. *Starts after: S2-DB-01; S2-DB-02*	US-25	Build	Vishva Vatalia	Duplicate budget for the same category/period returns 409; status endpoint returns a level per budget; tests pass.
S2-BE-01 Split calculation service services/splitting.py: compute_splits(total_minor, split_type, participants, exact_amounts, percents_bp, parts) for equal, exact, percentage (basis points, must total 10000) and parts (positive integers, e.g. 2:1:1). Uses money.allocate so remainder paise go one by one to participants in sorted user_id order. Pure functions. *Starts after: S0-DB-04*	US-18, US-19	Build	Jay Vaghela	100% branch coverage; property: result always sums to total; clear errors for invalid input.
S2-BE-02 Group expense endpoints GET/POST /groups/{id}/expenses, PATCH/DELETE /groups/{id}/expenses/{expense_id}. Accepts payers[] (MVP UI sends one payer) and participants; default split is equal among all current members; all payers/participants must be current members; writes expense, payers and splits in one transaction and records group_activity; only creator or admin can edit/delete. Contract stub on day 2. *Starts after: S2-BE-01; S1-BE-05; S1-DB-09*	US-18, US-19, US-23	Build	Jay Vaghela	Non-member participant returns 422; edits recompute splits; delete removes splits and payers; activity row written for add/edit/delete; tests pass.
S2-BE-03 Deploy backend to HTTPS staging Deploy to Render (free web service) from main: build/start commands, env vars, alembic upgrade head on deploy, /api/v1/health as health check. Document in docs/setup/deployment.md. *Starts after: S2-DB-05*	—	Build	Jay Vaghela	https://<app>.onrender.com/api/v1/health returns ok with db ok; secrets only in the dashboard; auto-redeploy on merge to main.

Task	Story	Type	Owner	Done when
S2-BE-04 Balances, debt simplification and settlements services/balances.py: net = paid (expense_payers) - owed (expense_splits) + settlements sent - settlements received, including former members; simplify_debts() greedy largest-debtor/largest-creditor matching (at most n-1 transfers). GET /groups/{id}/balances, GET /balances (You owe X / Y owes you totals), GET/POST /groups/{id}/settlements (any amount > 0, partial allowed, both users members or former members; one feature for US-10 AC4 and US-20); POST /groups/{id}/reminders (creditor reminds a debtor, at most once a day per pair, creates a payment_due notification); records activity. *Starts after: S2-BE-02; S2-DB-01; S2-DB-04*	US-10, US-20	Build	Yagnik Arvadiya	Sum of all net balances in a group is always 0 (tested); the ROADMAP.md §8.2 worked example is a test; a partial settlement reduces the balance by exactly its amount.
S2-BE-06 Invites, join, leave and remove member POST /groups/{id}/invites (open link/code or for an email), DELETE /groups/{id}/invites/{invite_id}, GET /invites (pending invites for my email), POST /invites/{code}/accept and /decline, POST /groups/{id}/leave (allowed with unsettled balance; returns balance so the app warns first), DELETE /groups/{id}/members/{user_id} (admin; the member's balance is shown to the admin first). Leaving/removal sets left_at and never deletes balances; all actions record activity and notify. *Starts after: S1-BE-05; S1-DB-09*	US-17, US-22	Build	Vivek Dodiya	Expired/revoked codes rejected; accepting twice is harmless; leaving keeps past expenses visible; only admins remove members; tests pass.

Database

Task	Story	Type	Owner	Done when
S2-DB-01 Models and migration: settlements, budgets settlements (from <> to check, amount > 0) and budgets (period check, limit > 0, alert_threshold_pct 1-100, source manual/ai_suggested, unique user/category/period with NULLS NOT DISTINCT). Merge by Wed 4 Nov. *Starts after: S1-DB-06*	US-20, US-25	Build	Vishva Vatalia	Migration applies cleanly; invalid rows rejected (tests).
S2-DB-02 Budget status query repositories/budgets.py: budget_status(user_id, today) returns for each active budget: period_start, period_end (IST, week starts Monday), spent_minor via user_spending, percent_used, level ok/warning/exceeded. *Starts after: S1-DB-07; S2-DB-01*	US-25	Build	Vishva Vatalia	Correct for weekly and monthly, category and overall budgets; tests at period boundaries.

Task	Story	Type	Owner	Done when
S2-DB-03 Analytics queries and endpoints repositories/analytics.py + router: GET /analytics/summary (period total, previous period, change %), /analytics/by-category (date range), /analytics/trend (granularity day/week/month, cumulative=true for a running total). All based on user_spending, so they update as soon as expenses change. *Starts after: S1-DB-07*	US-12	Build	Vishwa Patel	Endpoints documented; totals match a hand-calculated scenario; empty data returns zeros, not errors.
S2-DB-08 Analytics query tests Fixture-based tests: mixed personal/group expenses, multiple months, category filter, cumulative trend, user with no data. *Starts after: S2-DB-03*	US-12	Build	Vishwa Patel	At least 10 tests pass in CI.
S2-DB-04 Models and migration: recurring_rules, notifications, receipts, ai_usage_log Tables per ARCHITECTURE.md: recurring_rules (daily/weekly/monthly, status active/paused/cancelled, upcoming_notified_for), notifications (type list, data JSONB), receipts (image BYTEA, size check, expense_id FK ON DELETE CASCADE), ai_usage_log; indexes. Merge by Fri 6 Nov. *Starts after: S1-DB-06*	US-14, US-16	Build	Harshul Parikh	Migration applies cleanly; deleting an expense deletes its receipt row (test).
S2-DB-05 Provision cloud PostgreSQL and backup script Create a free managed Postgres (Neon recommended) for staging/production, connection string only in hosting env vars, run migrations, write scripts/backup_db.sh (pg_dump custom format, dated filename) and restore instructions. *Starts after: S2-DB-04*	US-34	Build	Harshul Parikh	Staging backend connects over SSL; a backup is produced and restored into a local DB successfully.
S2-DB-07 Split and payer integrity guard Service-level check that sum(share_minor) and sum(paid_minor) both equal amount_minor for every group expense, inside the same transaction; scripts/check_integrity.py reports violations (splits, payers, settlement members, negative amounts). *Starts after: S2-BE-01*	—	Build	Harshul Parikh	Mismatched splits or payers fail with 422; integrity script prints zero violations on seed data.

Gen-AI

Task	Story	Type	Owner	Done when
S2-AI-01 AI quota and usage logging ai/quota.py: per-user daily limits per feature from env (e.g. categorize 50, receipt 10, budget_goals 5, ask 20), log every call to ai_usage_log, global AI_ENABLED switch. Over the limit returns 429 and the app falls back to the manual flow. *Starts after: S2-DB-04; S1-AI-01*	—	Build	Yagnik Arvadiya	Limits configurable without code changes; usage rows written; 429 tested.

Task	Story	Type	Owner	Done when
S2-AI-02 AI budget goal suggestions endpoint GET /ai/budget-suggestions: compute the last 3 months of spending per category (no names/emails sent), LLM returns JSON [{category, suggested_monthly_limit_minor, reason}] for realistic goals; validate (category exists, limit > 0, within 50-150% of the 3-month average); rule-based fallback (average rounded to the nearest ₹100) when AI fails; cache per user per day. *Starts after: S2-DB-03; S2-DB-01; S2-AI-01*	US-25	Build	Aryan Panara	Always returns valid JSON; fallback works with AI disabled; no PII in the prompt (unit test); suggestions can be accepted through POST /budgets with source ai_suggested.
S2-AI-03 Bill scan endpoint with image storage POST /receipts/scan (multipart JPEG/PNG, max 5 MB): validates the file, checks quota, stores the image in receipts, calls ai/receipt.py, saves the extraction and returns receipt_id + fields; GET /receipts/{id}/image for the owner (or group members for group expenses); unlinked receipts older than 24 h are purged. *Starts after: S1-AI-03; S2-DB-04; S2-AI-01*	US-16	Build	Parthiv Panchal	Wrong type/size returns 415/413; AI failure still returns receipt_id with status failed so the user can type values; image deleted when its expense is deleted; tests with FakeProvider.

Testing

Task	Story	Type	Owner	Done when
S2-QA-01 Splitting and balances tests including property-based tests/unit/test_splitting.py and test_balances.py with hypothesis: random totals and participants always sum to total for every split type; random expense and settlement sets keep group net sum at 0; simplify_debts produces at most n-1 transfers that settle everything; partial settlements; former members. *Starts after: S2-BE-01; S2-BE-04*	US-19, US-20	Build	Aryan Panara	Property tests run in CI under 30 s; at least 15 example-based edge cases (1 paisa, 1 participant, payer not participating, parts 2:1:1).
S2-QA-02 API tests: groups, invites, group expenses, settlements, budgets Membership authorization (non-member gets 404 on every group endpoint), invite accept/decline/expiry, leave with balance, admin-only remove, participant validation, edit/delete permissions, settlement validation, budget CRUD and status levels. *Starts after: S2-BE-02; S2-BE-04; S2-BE-05; S2-BE-06*	US-17, US-18, US-20, US-22, US-23, US-25	Build	Parthiv Panchal	At least 25 test cases pass in CI; every new endpoint has an authorization test.
S2-QA-03 Device and browser session on HTTPS staging Point the app at the staging HTTPS URL, run Sprint 1 and 2 manual cases on 2 phones and Chrome, file and triage bugs with severity labels. *Starts after: S2-BE-03*	—	Research and process	Parthiv Panchal	Results committed; all S1/S2 severity bugs triaged before the sprint review.

Other planned work

Key dates: Wednesday 4 November, the settlements and budgets migration; Thursday 5 November, the split service and the contract pull requests for shared expenses, balances, invites, budgets and analytics; Friday 6 November, the recurring, notifications and receipts migration; Tuesday 10 November, HTTPS staging live. From then on all device testing points to staging.

Testing focus

Property tests for splits, payers and balances; authorization tests for groups, invites and budgets; a session on HTTPS staging.

Sprint review

- **Demonstration.** Against HTTPS staging: create “Goa Trip”, join from a second phone through a shared link and accept an email invite in-app; add a dinner split equally, a hotel split by exact amounts and fuel split by parts (2:1:1); view balances and record a partial settlement; edit an expense and show the group history; a member leaves after the warning and their history stays; search with combined filters and the spending-over-time chart on the website; budget suggestions and bill extraction in Swagger.

Exit criteria

Sprint 2 is complete only when all of the following are true:

1. Property tests prove that splits and payers always sum to the total and that group balances always sum to 0
2. Every group endpoint has a non-member authorization test
3. Staging is reachable over HTTPS from a phone on mobile data

7. Sprint 3: Smart and automatic

Planned dates	17 – 25 November 2026
Sprint review	Wednesday 25 November 2026
Scrum Master	Aryan Panara
Story points	67 (55 story + 12 setup and testing)
Sprint goal	Users can scan a bill to create an expense, set budget goals with AI and get alerts and congratulations, chat with the AI about their spending, automate recurring expenses with a day-before notice, manage group details and co-admins, change their password, and see everything in a notification centre, in an installable APK and on a live website.

User stories in this sprint

ID	Title	Priority	Points	Owner
US-03	Password Visibility Toggle	Must	1	Vivek Dodiya
US-06	Change Password	Must	3	Jay Vaghela, with Aryan Panara, Vivek Dodiya and Parthiv Panchal
US-10	Track Money Owed	Must	1	Vishwa Patel, with Vivek Dodiya
US-12	Total Spending Over Time	Must	1	Vishwa Patel
US-14	Recurring Expenses	Must	9	Harshul Parikh, with Vivek Dodiya and Parthiv Panchal
US-16	Scan Bill to Create Expense	Must	7	Ved Bhoraniya, with Parthiv Panchal and Aryan Panara
US-19	Unequal Split	Must	1	Aryan Panara
US-21	Manage Group Details and Roles	Must	6	Vivek Dodiya, with Vishva Vatalia, Vishwa Patel and Parthiv Panchal
US-23	Edit or Delete Group Expense	Must	1	Vishwa Patel, with Vivek Dodiya
US-24	AI Chat Assistant	Must	6	Yagnik Arvadiya, with Ved Bhoraniya and Parthiv Panchal
US-25	AI Goal Setting	Must	9	Riya Soni, with Ved Bhoraniya, Vishva Vatalia, Vishwa Patel, Aryan Panara, Parthiv Panchal and Vivek Dodiya
US-26	AI Category Suggestion	Must	1	Aryan Panara
US-32	Performance and Responsiveness	Must	2	Harshul Parikh, with Parthiv Panchal
US-33	Privacy and Security	Must	3	Jay Vaghela
US-34	Data Integrity	Must	2	Vishva Vatalia
US-35	Scalability and Availability	Must	2	Jay Vaghela, with Harshul Parikh
	Total		**55**	

Points are the backlog estimates of this sprint's tasks for each story; a task that serves several stories is split between them. "Groundwork" stories are delivered in a later sprint. The other 12 points are setup and testing tasks with no story.

Work by domain

Frontend

Task	Story	Type	Owner	Done when
S3-FE-01 Bill scan flow Scan bill button on Add Expense: camera or gallery (expo-image-picker; file picker on web), compress with expo-image-manipulator, upload with progress, review screen with the extracted fields pre-filled and editable plus the image, Save creates the expense linked to receipt_id; on failure open the manual form with the image visible. Expense details show a View bill button. *Starts after: S2-AI-03; S1-FE-05*	US-16	Build	Ved Bhoraniya	Never saves without user confirmation; works with AI disabled; bill image viewable from expense details; tested on 2 phones and Chrome with real bills.
S3-FE-02 AI budget goal suggestions UI On the budgets screen, Suggest goals with AI shows the suggested monthly limit and reason per category; the user can accept, edit the amount, or dismiss each suggestion; accepted ones are saved as budgets. Label AI-generated. *Starts after: S2-AI-02; S3-FE-10*	US-25	Build	Ved Bhoraniya	Shows rule-based suggestions when AI is unavailable; nothing is saved without the user accepting.
S3-FE-05 Ask AI chat screen Chat-style screen: question input, suggested questions (How much did I spend on food last week?), text answers with the numbers used, fallback suggestions when it cannot answer, disclaimer that it is not financial advice. Also an About AI & privacy note, linked from every AI feature, explaining what is sent to the external AI service. *Starts after: S3-AI-01*	US-24	Build	Ved Bhoraniya	Answers come from the user's own data; out-of-scope questions show fallback suggestions; disabled state when the AI quota is exceeded.
S3-FE-03 Recurring expenses screens Mark an expense as recurring (daily, weekly, monthly) from the add form or a Recurring screen; list with next date, pause/resume and cancel; call /recurring/run-due in the background after the first screen appears (never blocking start-up). *Starts after: S3-DB-01; S1-FE-05*	US-14	Build	Vivek Dodiya	Paused or cancelled rules add nothing; generated expenses appear in history marked as recurring; the day-before notification is visible.
S3-FE-04 Notifications screen and badge Bell icon with unread badge (unread-count on focus and every 30 s), notifications list with type icons, tap marks read and opens the related screen (budget, group, expense, recurring), Mark all read. *Starts after: S3-BE-02*	US-10, US-14, US-21, US-23, US-25	Build	Vivek Dodiya	Badge updates after reading; navigation targets correct; empty state.
S3-FE-08 Group settings and roles screen Admin-only settings: edit name and description, promote a member to co-admin or demote, remove member, delete group (blocked with a message while balances are unsettled). Members see a read-only view. *Starts after: S3-BE-07*	US-21	Build	Vivek Dodiya	Non-admins cannot see admin actions; changes appear for other members and they receive a notification.

Task	Story	Type	Owner	Done when
S3-FE-09 Change password screen Form with current password, new password and confirmation; PasswordField eye toggles; real-time strength feedback (length, letters, numbers, symbols); in-app confirmation message on success. *Starts after: S3-BE-05; S1-FE-01*	US-03, US-06	Build	Vivek Dodiya	Weak or mismatched passwords cannot be submitted; wrong current password shows an error; other devices are logged out after a change.
S3-FE-10 Budgets screens Budgets list with progress bars coloured by level (ok/warning/exceeded), create/edit budget (category or overall, weekly or monthly, limit, alert threshold), delete. *Starts after: S2-BE-05; S1-FE-01*	US-25	Build	Riya Soni	Progress matches /budgets/status; validation mirrors the backend; works on Android and web.
S3-OPS-01 EAS build config and first preview APK eas.json with development, preview (buildType apk, staging HTTPS URL) and production profiles; app.json android.package, versionCode, icon and splash placeholders; build the preview APK by Fri 20 Nov and share the install link. *Starts after: S2-BE-03*	—	Build	Riya Soni	APK installs on 2+ phones and talks to the HTTPS backend; steps in docs/setup/release.md.
S3-OPS-02 Website build and deployment (staging) app.json web output single, npx expo export --platform web, deploy dist/ to Vercel or Netlify with an SPA rewrite so links like /groups/12 and /join/CODE load, EXPO_PUBLIC_API_URL set to the staging backend, auto-deploy from main; add the website origin to backend CORS_ORIGINS. Live by Fri 20 Nov. *Starts after: S2-BE-03; S1-FE-07*	—	Build	Riya Soni	Website loads over HTTPS; login and core flows work in Chrome and one other browser; refreshing a deep link does not 404; steps in docs/setup/release.md.

Backend

Task	Story	Type	Owner	Done when
S3-BE-02 Notifications API GET /notifications (paginated), GET /notifications/unread-count, POST /notifications/{id}/read, POST /notifications/read-all; owner-only. *Starts after: S2-DB-04*	—	Build	Jay Vaghela	Endpoints tested; another user's notification returns 404.
S3-BE-03 Security hardening and production logging Rate limit login/register (slowapi, e.g. 5/min per IP), request body size limits, CORS restricted to the website origin in production, authorization review of every router, no stack traces or secrets in responses, JWT secret length check, structured request logs without bodies or passwords. *Starts after: S2-BE-04*	US-33	Build	Jay Vaghela	Authorization checklist in docs/testing/security-checklist.md all ticked; brute-force login test returns 429.
S3-BE-05 Change password endpoint POST /auth/change-password: verifies the current password, enforces strength rules (min 8 chars, letters and numbers), hashes the new one, revokes all other sessions, creates an in-app confirmation notification. *Starts after: S1-BE-01*	US-06	Build	Jay Vaghela	Wrong current password returns 400; weak password returns 422; other sessions cannot refresh afterwards; tests pass.

Task	Story	Type	Owner	Done when
S3-BE-06 Maintenance announcement script scripts/announce.py creates a system notification for all active users with the planned maintenance window; documented in docs/setup/deployment.md. *Starts after: S3-BE-02*	US-35	Build	Jay Vaghela	Running the script on staging shows the notice to every user once.
S3-BE-07 Group details and roles API PATCH /groups/{id} (name, description), PATCH /groups/{id}/members/{user_id} (promote to co-admin = admin role, demote; the last admin cannot be demoted), DELETE /groups/{id} (admin; blocked with 409 while any balance is non-zero); record activity and notify members. *Starts after: S1-BE-05; S1-DB-09*	US-21	Build	Vishva Vatalia	Only admins can perform these actions (tested); last-admin rule enforced; members notified.
S3-BE-01 Notification triggers services/notifications.py: budget_warning at the threshold and budget_exceeded at 100% (once per budget per period per level); goal_achieved when a period ends under the limit (created the first time status is checked after the period ends); payment_due to participants when a shared expense is added; settlement_received; group_expense_changed to members on edit/delete; group_updated and role_changed; group_invite for email invites. *Starts after: S2-DB-02; S2-DB-04; S2-BE-04*	US-10, US-21, US-23, US-25	Build	Vishwa Patel	No duplicate notifications for the same event (tested); notifications created in the same transaction as the change.

Database

Task	Story	Type	Owner	Done when
S3-DB-01 Recurring rules service and API GET/POST /recurring, PATCH/DELETE /recurring/{id} (edit, pause/resume, cancel). One idempotent service, run for all users daily at 06:00 IST by a GitHub Actions scheduled workflow calling POST /internal/recurring/run (protected by a secret header), and for the current user by POST /recurring/run-due when the app opens (fallback). It sends a recurring_upcoming notification the day before next_run_date (deduped by upcoming_notified_for), adds every due expense (expense_date <= today), advances next_run_date month-end safely (31 Jan to 28 Feb), idempotent via the unique (recurring_rule_id, expense_date) index, creates recurring_added notification. *Starts after: S2-DB-04*	US-14	Build	Harshul Parikh	Calling run-due twice creates no duplicates; the day-before notice is sent once; paused or cancelled rules add nothing; tests pass.
S3-DB-02 Performance review Run EXPLAIN ANALYZE on expense list/search, user_spending, analytics and balances with 5,000 seeded expenses; add missing indexes via migration; measure dashboard API time on staging. *Starts after: S2-DB-03*	US-32, US-35	Build	Harshul Parikh	No main query above 200 ms locally on seeded data; results in docs/design/db-performance.md.

Task	Story	Type	Owner	Done when
S3-DB-03 Final demo dataset scripts/seed_demo.py: 4 demo users, 3 months of realistic expenses with payment modes, 2 groups with mixed splits, settlements and activity, budgets near and over limits, 2 recurring rules, so every feature has data in the demo. *Starts after: S3-DB-01; S2-DB-07*	—	Build	Vishwa Patel	Running the script on an empty DB fills every dashboard widget; credentials documented privately.
S3-DB-04 Analytics edge cases Fix and test: months with no data, weekly granularity across a month boundary, deleted categories, very large amounts (up to 10^9 paise). *Starts after: S2-DB-08*	US-12	Build	Vishwa Patel	Edge case tests pass; no 500 errors on empty or extreme data.
S3-DB-05 Budget period edge cases Tests and fixes: week starting Monday, IST midnight boundaries, month change, editing an expense date that moves spending between periods. *Starts after: S2-DB-02*	US-25	Build	Vishva Vatalia	All boundary tests pass; behaviour documented in ARCHITECTURE.md.
S3-DB-06 Backup and restore drill Run scripts/backup_db.sh against the cloud DB, restore into a fresh local DB, verify row counts with a script; weekly backup owner until the presentation. *Starts after: S2-DB-05*	US-34	Build	Vishva Vatalia	Drill log in docs/setup/backup.md with date, file and verification output.

Gen-AI

Task	Story	Type	Owner	Done when
S3-AI-01 Ask AI endpoint POST /ai/ask: step 1 the LLM maps the question to an intent from a fixed list (spend_total, spend_by_category, top_merchants, budget_status, group_balance, compare_periods) with parameters via structured output; step 2 the backend runs the whitelisted query (never LLM-written SQL); step 3 the answer is phrased using only the returned numbers. Unanswerable questions return fallback suggestions. *Starts after: S2-DB-03; S2-AI-01*	US-24	Build	Yagnik Arvadiya	Never executes free-form SQL; numbers in answers match the query result; out-of-scope and prompt-injection tests pass with FakeProvider.
S3-AI-02 AI resilience and privacy review Verify timeouts, single retry, fallbacks and quotas on all AI endpoints; switch provider/model via env only; unit test asserting prompts contain no email or full name; document in docs/ai/README.md. *Starts after: S2-AI-02; S2-AI-03*	—	Build	Yagnik Arvadiya	Each AI feature works with AI_ENABLED=false; privacy test passes; provider swap documented.
S3-AI-03 AI evaluation and prompt tuning Run categorization and bill extraction evals on the golden sets, tune prompts (few-shot, Indian merchants), record before/after accuracy and latency in docs/ai/eval-sprint3.md. Targets: categorization >= 80%, bill total exact match >= 80%. *Starts after: S1-AI-02; S2-AI-03*	US-16, US-26	Build	Aryan Panara	Eval report committed; prompt changes merged via PR with eval numbers in the description.

Task	Story	Type	Owner	Done when
S3-AI-04 Bill extraction tuning Test 15+ real bills (thermal, restaurant with GST, UPI screenshots); improve prompt and post-processing (DD/MM/YYYY dates, rupee symbols, totals vs subtotals). *Starts after: S2-AI-03*	US-16	Build	Parthiv Panchal	Measured improvement documented; regressions covered by saved fixtures.

Testing

Task	Story	Type	Owner	Done when
S3-QA-01 Full regression test suite (manual) Complete docs/testing/test-cases.md: at least one case per acceptance criterion of every story planned for the release (see ROADMAP.md §2.2), including negative cases and slow network, each tagged Android/web. *Starts after: S2-QA-03*	—	Build	Aryan Panara	Every release story has cases; reviewed by Parthiv.
S3-QA-02 Component tests: split editor, budgets, password strength Jest tests for split editor validation (exact/percent/parts mismatch disables Save, remainder preview), budget progress levels, and change-password strength feedback. *Starts after: S2-FE-02; S3-FE-10; S3-FE-09*	US-06, US-19, US-25	Build	Aryan Panara	Tests pass in mobile CI.
S3-QA-03 API tests: recurring, notifications, receipts, AI, roles Tests with FakeProvider for bill scan, budget suggestions and ask (success, failure, 429); recurring idempotency, day-before notice and month-end; notification dedupe and authorization; role changes and last-admin rule; change password. *Starts after: S3-DB-01; S3-BE-02; S3-BE-07; S3-AI-01*	US-06, US-14, US-16, US-21, US-24, US-25	Build	Parthiv Panchal	At least 25 test cases pass in CI.
S3-QA-04 Preview APK and website testing Install the preview APK on 2+ Android phones (different versions/screen sizes) and open the staging website in Chrome plus one other browser at desktop and phone widths; run the smoke suite on each; file bugs. *Starts after: S3-OPS-01; S3-OPS-02*	US-32	Research and process	Parthiv Panchal	Device/browser matrix committed; APK-specific (cleartext, permissions, camera) and web-specific (layout, file upload, storage) issues identified.

Other planned work

Key dates: Wednesday 18 November, the contract pull requests for notifications, recurring, roles and Ask AI; Friday 20 November, the preview APK, the staging website and the budgets screens; Monday 23 November, all feature pull requests open. Feature freeze Wednesday 25 November, 23:59; after it only bug fixes, tests and documents are merged.

Testing focus

API tests for recurring, notifications, AI and roles; the full regression suite; APK and website testing; the AI evaluation.

Sprint review

- **Demonstration.** On the APK with the laptop closed: scan a real restaurant bill, edit, confirm and view the bill from the expense; accept two AI budget goals and cross 80% to get a warning; ask the chat “How much

did I spend on food last week?"; a monthly rent rule sends a "due tomorrow" notice and then adds the expense; promote a co-admin; change the password and see the other device logged out; then the same data on the website, with a bill uploaded from the laptop.

Exit criteria

Sprint 3 is complete only when all of the following are true:

1. Every acceptance criterion in the release scope has a passing test case, except the deferred criteria D1–D11 in ROADMAP.md §3.5
2. AI features work with AI_ENABLED=false
3. The preview APK is verified on 2 or more phones and the website in 2 browsers

8. Sprint 4: Ship it

Planned dates	26 – 30 November 2026
Sprint review	Monday 30 November 2026 (final presentation)
Scrum Master	Yagnik Arvadiya
Story points	32
Sprint goal	Deliver a stable v1.0.0 APK and website on the production HTTPS backend, with a full regression pass, a rehearsed demo, and every member ready for their viva.

User stories in this sprint

Release sprint: no new user stories. The regression covers all 28 release stories.

Work by domain

Frontend

Task	Story	Type	Owner	Done when
S4-FE-01 Release bug fixes: auth, groups, recurring, notifications Fix bugs assigned from regression in these areas; write mobile/README.md (run, build, env vars). *Starts after: S4-QA-02*	—	Build	Vivek Dodiya	All assigned S1/S2 bugs closed with linked PRs; README reviewed.
S4-FE-02 Release bug fixes: expenses, bills, analytics, AI UI Fix assigned bugs; verify AI fallbacks with AI_ENABLED=false on staging and in airplane mode; prepare the AI part of the demo script. *Starts after: S4-QA-01*	—	Build	Ved Bhoraniya	All assigned S1/S2 bugs closed; demo script in docs/sprints/sprint-4 /demo-script.md.
S4-FE-03 Production APK and website release Final icon, splash and favicon, version 1.0.0, production profile pointing to the production HTTPS backend; build the release-candidate APK and website on Thu 26 Nov and final versions on Sun 29 Nov; attach the APK to the GitHub Release and promote the website to production. *Starts after: S3-OPS-01; S3-OPS-02*	—	Build	Riya Soni	APK installs on 3 phones; production website works in 2 browsers; GitHub Release v1.0.0 has the APK, website link and changelog.

Backend

Task	Story	Type	Owner	Done when
S4-BE-01 Production deploy and smoke test script Final deploy, tag v1.0.0, scripts/smoke_test.py hitting production health, register, login, create expense and group balances; keep-warm and warm-up instructions for the demo. *Starts after: S4-FE-03*	—	Build	Jay Vaghela	Smoke script passes against production; tag created.

Database

Task	Story	Type	Owner	Done when
S4-DB-01 Production data integrity check Run scripts/check_integrity.py on production (split and payer sums, settlement members, orphan receipts, negative amounts) and fix issues via migration or script. *Starts after: S4-BE-01*	US-34	Build	Harshul Parikh	Integrity report shows zero violations; output attached to the issue.
S4-DB-02 Load demo data on production Run seed_demo.py on production with demo accounts, verify every screen has data, rehearse the demo login. *Starts after: S3-DB-03; S4-BE-01*	—	Build	Vishwa Patel	Demo accounts work on the final APK and website.
S4-DB-03 Final backup and DB documentation Final backup before the presentation, restore verification, update the ER diagram and schema docs to match the last migration. *Starts after: S4-BE-01*	US-34	Build	Vishva Vatalia	Backup verified; ARCHITECTURE.md schema matches alembic head.

Gen-AI

Task	Story	Type	Owner	Done when
S4-AI-01 AI release checks and usage report Fix AI bugs from regression, freeze prompt versions, usage report from ai_usage_log, verify quotas in production, rotate any key that was ever shared. *Starts after: S4-QA-01*	—	Build	Yagnik Arvadiya	Report in docs/ai/final-report.md; production AI and fallbacks verified.

Testing

Task	Story	Type	Owner	Done when
S4-QA-01 Final regression: auth, personal expenses, AI, non-functional Run regression for US-01, US-02, US-03, US-05, US-06, US-07, US-08, US-09, US-12, US-13, US-15, US-24, US-25, US-26 and US-32 to US-35 on the release-candidate APK and production website; retest fixed bugs; add an automated test for each fixed backend bug. *Starts after: S3-QA-01; S4-FE-03*	—	Build	Aryan Panara	Section of docs/testing/final-test-report.md complete; zero open S1/S2 bugs in these stories.
S4-QA-02 Final regression: groups, balances, bills, recurring Run regression for US-10, US-14 and US-16 to US-23 on the release-candidate APK and production website; retest fixed bugs; add automated regression tests. *Starts after: S3-QA-01; S4-FE-03*	—	Build	Parthiv Panchal	Final test report section complete; zero open S1/S2 bugs in these stories.

DevOps

Task	Story	Type	Owner	Done when
S4-DOC-01 Final presentation and viva preparation Final presentation and demo (app + website), backup demo video. Each member finalizes docs/contributions/<name>.md (what they built, key PRs, decisions and why, bugs fixed, what they would improve) and does a 15-minute mock viva with their reviewer pair (ROADMAP.md §11.4). *Starts after: S4-QA-01; S4-QA-02*	—	Research and process	Whole team	Presentation ready; every contribution file committed; every member has done a mock viva by Sun 29 Nov.

Other planned work

Thursday 26 November: release-candidate APK and website; the full regression starts. Friday 27 – Saturday 28 November: bug fixing, each fix with a regression test; production deploy, demo data and integrity check on Saturday 28 November. Code freeze Sunday 29 November, 20:00: final APK in GitHub Release v1.0.0, production website deployed, final backup, mock vivas done.

Testing focus

Full regression on the release candidate and the final test report.

Sprint review

- **Demonstration.** Final presentation on Monday 30 November: the backend is warmed up, the demo script is run on the app and the website, then the viva.

Exit criteria

Sprint 4 is complete only when all of the following are true:

1. There are zero open S1/S2 bugs
2. The final test report is committed
3. The APK is in GitHub Release v1.0.0 and the production website is live
4. Every member has a contribution log and has done a mock viva

9. Stretch backlog

Stretch items are not planned into a sprint. They are taken in at sprint planning, Should items first, only when that sprint's Must work is on track, and never after the feature freeze.

ID	Item	Stories	Priority	Points	Owners
ST-01	Forgot password via email	US-04	Should	5	Jay Vaghela, Vivek Dodiya
ST-02	Active sessions management	US-05 AC3	Should	2	Jay Vaghela, Vivek Dodiya
ST-03	Income entries and monthly savings	US-11	Should	5	Vishwa Patel, Ved Bhoraniya
ST-04	Quick-add frequent expenses	US-07 AC4	Should	2	Vishwa Patel, Ved Bhoraniya
ST-05	Multiple payers screen	US-19 AC3	Should	3	Jay Vaghela, Riya Soni
ST-06	Bills on shared expenses and PDF bills	US-18 AC5, US-16 AC1	Should	3	Parthiv Panchal, Ved Bhoraniya
ST-07	Group photo	US-21 AC1	Should	2	Vishva Vatalia, Riya Soni
ST-08	Email delivery for invites and confirmations	US-17 AC2	Should	2	Jay Vaghela
ST-09	Offline add and sync without duplicates	US-07 AC5, US-34 AC3	Could	5	Vivek Dodiya, Vishwa Patel
ST-10	Customer support	US-27, US-28, US-29	Could	8	Riya Soni, Jay Vaghela, Harshul Parikh
ST-11	System administration and usage overview	US-30, US-31	Could	5	Jay Vaghela, Vishva Vatalia, Riya Soni
ST-12	Optional two-factor authentication	US-33 AC3	Could	3	Jay Vaghela, Vivek Dodiya
ST-13	Push notifications	—	Could	3	Vivek Dodiya, Jay Vaghela
ST-14	Parental mode	US-36, US-37, US-38	Could	8	Yagnik Arvadiya, Vishwa Patel, Ved Bhoraniya
ST-15	Multiple currencies	US-30	Could	5	Jay Vaghela, Vishva Vatalia
ST-16	Notification preferences	US-39	Could	3	Vishwa Patel, Vivek Dodiya

10. Work summary by member

Number of tasks each member owns in each sprint, and the story points they own. Every member owns tasks in every sprint.

Member	Role	S0	S1	S2	S3	S4	Tasks	Points
Jay Vaghela	Backend Engineer	2	2	3	4	2	13	33
Vivek Dodiya	Frontend Engineer	3	2	2	4	2	13	35
Ved Bhoraniya	Frontend Engineer	2	4	2	3	2	13	31
Riya Soni	Frontend Engineer (UI/UX lead)	2	3	3	3	2	13	33
Yagnik Arvadiya	Gen-AI Engineer	1	2	2	2	2	9	29
Vishwa Patel	Database Engineer	2	3	2	3	2	12	30
Vishva Vatalia	Database Engineer	1	3	3	3	2	12	28
Harshul Parikh	Database Engineer	1	3	3	2	2	11	29
Aryan Panara	Gen-AI · Testing	3	3	2	3	2	13	31
Parthiv Panchal	Testing Engineer	2	3	3	3	2	13	31
Total		**19**	**28**	**25**	**30**	**20**	**122**	**310**

S4-DOC-01 (final presentation and viva preparation, 2 points) is shared by all 10 members and is counted in every member's row, so the member totals add up to 310 points and 122 tasks, while the backlog holds 113 tasks worth 292 points.